This implies that $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction,

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \cdots + \cos(\alpha+(n-1)\beta)$$

$$= \cos\left(\alpha + \frac{(n-1)\beta}{2}\right) \times \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$$

Example 4.70 Using the Mathematical induction, show that for any natural number n, with the assumption $i^2 = -1$,

$$(r(\cos\theta + i\sin\theta))^n = r^n(\cos n\theta + i\sin n\theta),$$

Solution:
Let,

$$P(n) = (r(\cos\theta + i\sin\theta))^n = r^n(\cos n\theta + i\sin n\theta).$$

Substituting the value of $n = 1$, in the statement we get,

$$P(1) = (r(\cos\theta + i\sin\theta))^1 = r(\cos\theta + i\sin\theta).$$

Hence, $P(1)$ is true. Let us assume that the statement is true for $n = k$. Then

$$(r(\cos\theta + i\sin\theta))^k = r^k(\cos k\theta + i\sin k\theta),$$

We need to show that $P(k+1)$ is true. Consider,

$$\begin{aligned}
P(k+1) &= (r(\cos\theta + i\sin\theta))^{k+1} \\
&= (r(\cos\theta + i\sin\theta))^k \times r(\cos\theta + i\sin\theta) \\
&= r^k(\cos k\theta + i\sin k\theta) \times r(\cos\theta + i\sin\theta) \\
&= r^{k+1} \times \big((\cos k\theta\cos\theta + i^2\sin k\theta\sin\theta) + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta)\big) \\
&= r^{k+1} \times ((\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta)) \\
&= r^{k+1} \times (\cos(k+1)\theta + i\sin(k+1)\theta).
\end{aligned}$$

This implies that $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction, for any natural number n,

$$(r(\cos\theta + i\sin\theta))^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

.



What we have proved in Example 4.70 is called the **Demoivre's theorem** for natural numbers, which will be studied in detail in the second year of Higher Secondary course.